Month 8 Review and Recovery Plan

ARES CONSORTIUM

Solis

THARSIS

Meridian

helix

Project Status

Four things have gone wrong at month 8. None is a failure of effort. All four follow from one unwritten document: the contract saying what crosses between the models, in what units, owned by whom.

The one Solis to Tharsis exchange is two months late, in the wrong format.
Nobody wrote down when it was due or what it should look like, so there was no format to be late in.

Tharsis and Helix disagree about what the interface variables mean and who owns them.
No partner was named as owner, so Helix picked meanings and built its tooling on them.

Meridian, the risk laboratory, says the interface cannot represent failure or uncertainty.
It needs every exchanged number to arrive with a stated uncertainty, and none of them do.

Nobody has shown the two models need to run together.
The two conversions between Tharsis's hourly simulation and Solis's multi-year plan belong to no partner.

Second-year funding depends on a demonstration, and there is no demonstration.

Ownership Gap

Five inputs and conversions the models need, and no partner produces. Of six possible partner pairs, only two state any exchange at all.

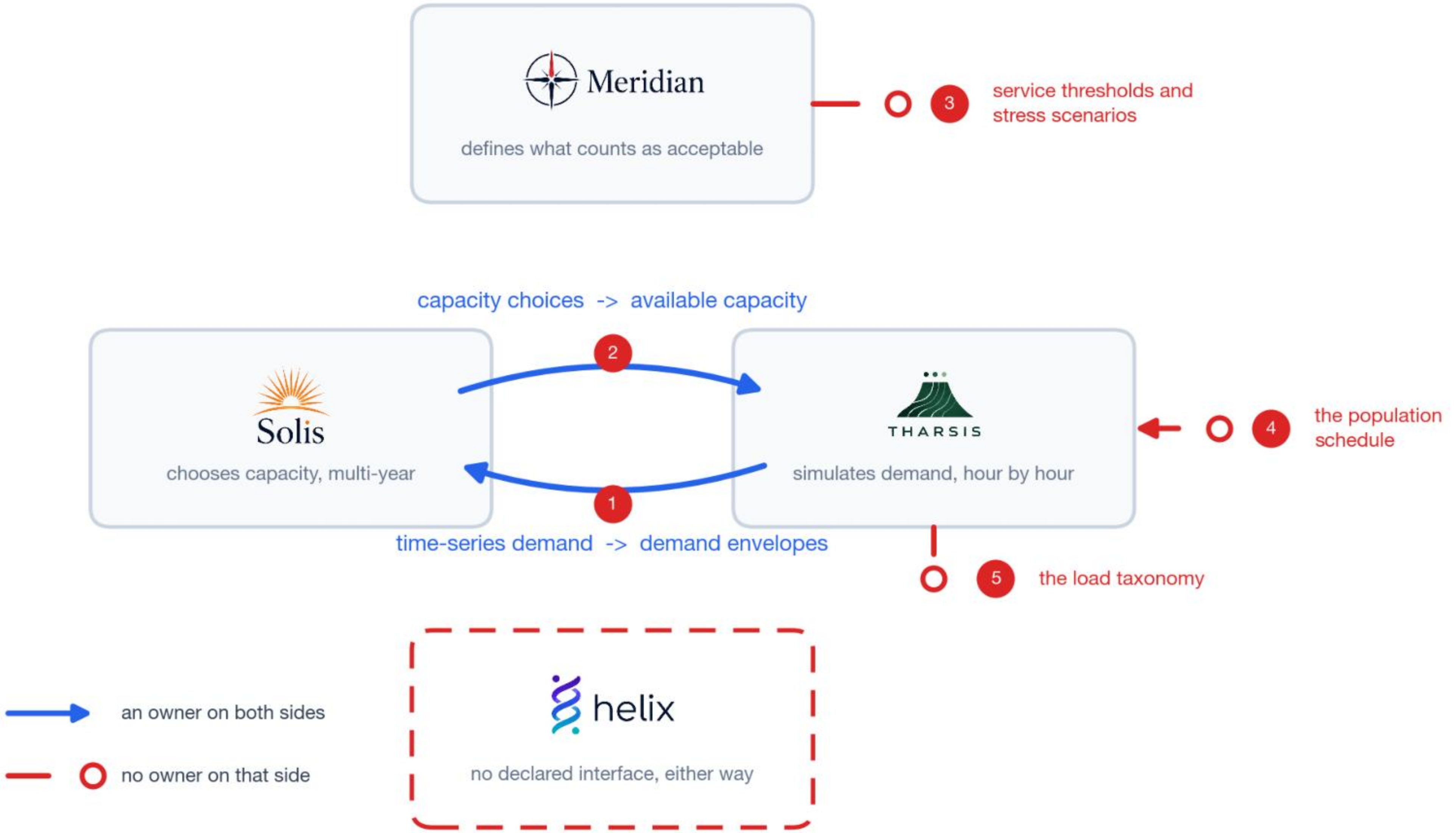
1. The aggregation operator.
Turns Tharsis's hourly demand into the bound Solis plans against. Nobody owns it.

2. The derate function.
Turns Solis's installed capacity into the available capacity Tharsis needs. Tharsis invents its own.

3. The service thresholds and stress scenarios.
Meridian produces both and routes them to nobody.

4. The population schedule.
Drives every demand profile. No partner produces it and nobody versions it.

5. The load taxonomy.
Which loads can be shed in a storm and which cannot. Undeclared.



Model Fidelity

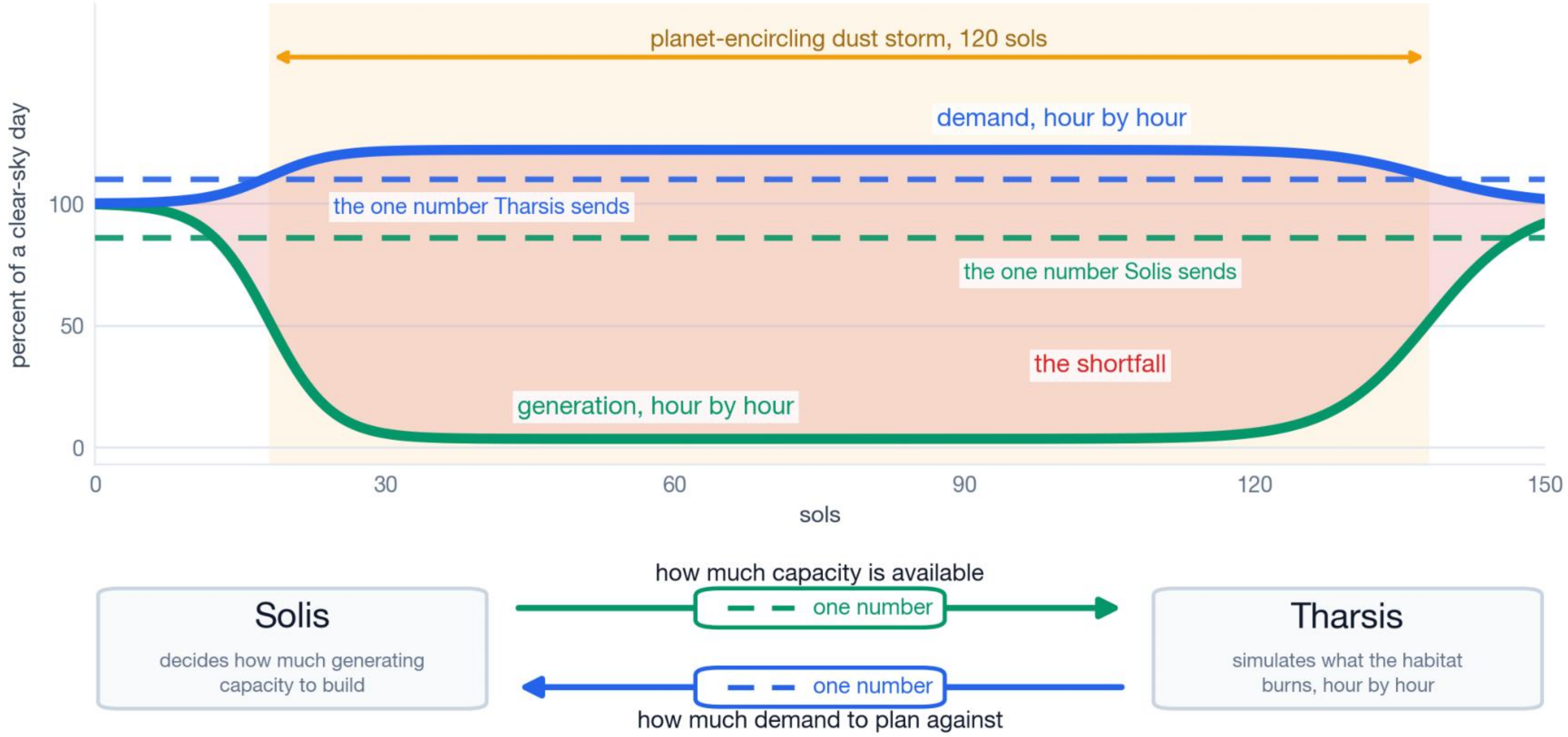
How good does this model have to be before an engineer will sign a habitat design on it, knowing people die if it is wrong?

A dust storm cuts generation and raises demand at the same time.
It darkens the arrays and cuts passive solar heat into the habitat, so heating load rises as generation falls.

The interface sends a single bound, so the two effects arrive separately.
Each model sees a worst case for its own side, never both at once.

Meridian's clause 4.2 already forbids this.
Where two stressors share a physical cause, the model shall not treat them as independent.

Somebody signs a habitat design on this model.
Too coarse and it is not weak evidence, it is no evidence, and Meridian has already said the interface cannot carry the claims made from it.



Delivery Approach

From week 2 the demonstration runs. Every week after that adds to a run that already works.

Week 2 runs end to end on crude inputs.
One sol, nothing polished. Every format mismatch fails while there is time.

Month 8, and it has never run end to end.
Nothing has ever been tested across the boundary.

Nothing is taken apart to add the next piece.
Worst case in week 6 is week 5's version.

Week 5 is buffer.
Nothing scheduled into it that cannot move out.

Week 5 also tests this recommendation.
Solis revises after seeing the stress run. If it barely changes its design, the feedback is weak and tight coupling was not needed.

Six Week Plan

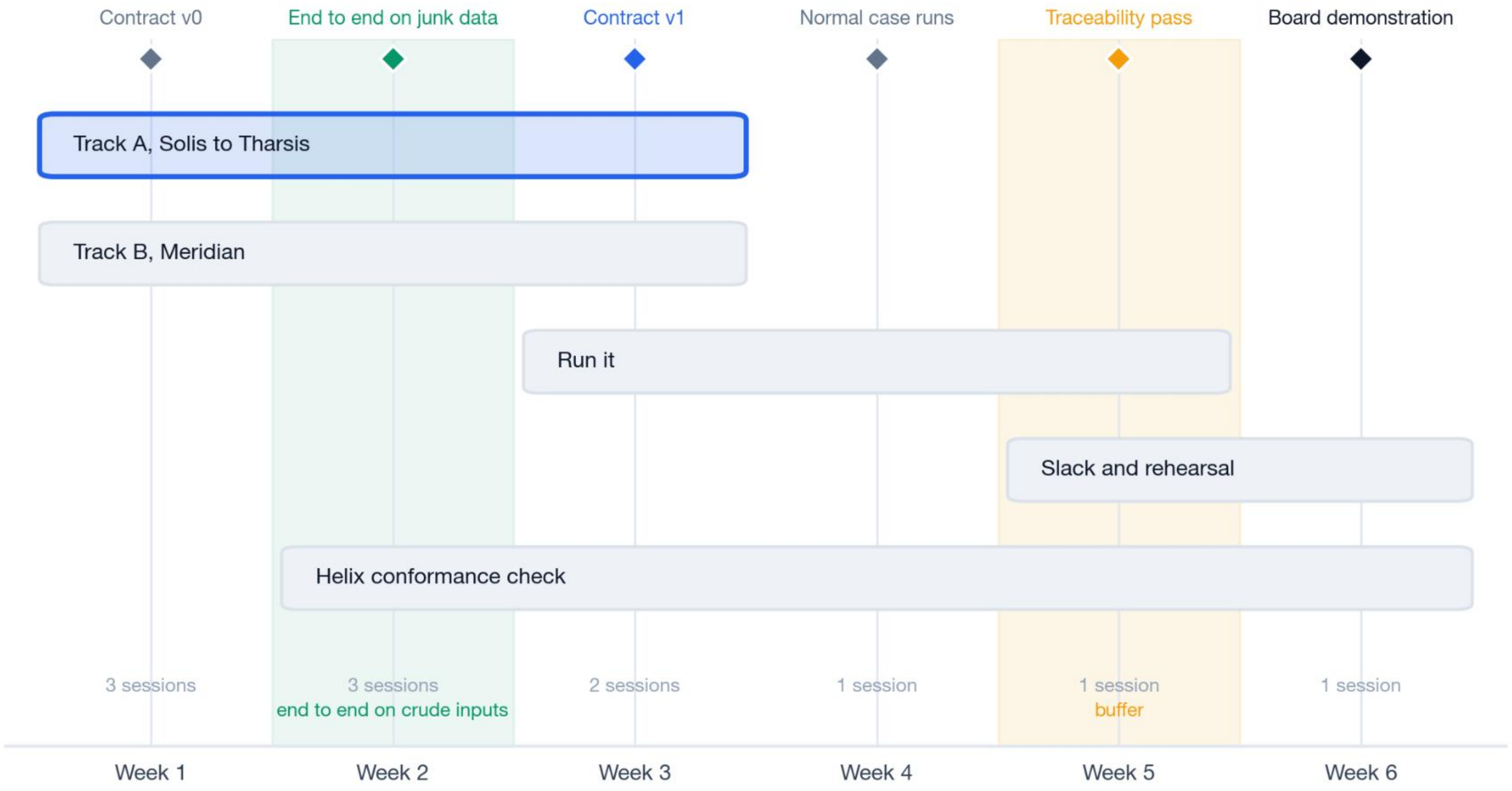
Solis to Tharsis is the critical path. Meridian runs alongside it. Week 5 absorbs a slip and week 6 rehearses.

Track A
is the critical path. The exchange that already failed.

Track B
runs from week 1, because Meridian sets what the contract has to carry.

Weeks 3 and 4.
Normal case, then the stress event, iterated.

Week 6.
Every number traced back to the model and contract that produced it. That is the deliverable.



Governance

Eleven sessions in six weeks. Helix sits in seven of them and holds not one ruling.

Five of the eleven fall in weeks 1 and 2
including the derate session, which is the one decision here that is hard to reverse.

Two to three touchpoints a week
for the coordinator.

Helix is in seven rooms and holds no rulings.
Present when the contract it built against is settled, never holding the decision.

Helix holds the only working transport code for the failed exchange.

Partner Alignment

Every partner is worse off under the status quo, and three do not yet know how.

Helix
built its tooling on a guess at what the variables mean. A written contract is the only outcome where that work survives. Approach first.

Meridian
has already asked for uncertainty on every exchanged number, in the form of an objection. Agree with it in public.

Solis
is judged on a derate Tharsis wrote and Solis never saw. Writing the derate down is how Solis gets a say in it.

Tharsis
carries blame for thresholds Meridian never sent and a schedule the programme never owned. Writing down who sends each of those moves the blame to them.

All four are currently blamed for something the contract never assigned to them. Week 1 is when that is easiest to say out loud.

Decisions Required

Four decisions. Three change only what the coordinator may ask for. The fourth narrows what week six delivers.

Endorse the diagnosis.
Five things the models need that no partner produces, all on the Solis to Tharsis boundary. Everything below follows from that.

Back the sequence.
Solis to Tharsis first, Meridian alongside. Nothing else starts until those two exchange successfully.

Give the coordinator standing to make each partner write its side down.
Units, calendars, owners and what crosses each boundary. Methods and code stay with the partner.

Accept a narrow demonstration.
One configuration, one stress event, fully traceable. Not a platform, and not four couplings. This is the decision that costs you something.

In return, at week six.
One configuration under a normal case and a material stress event, every number traceable to the model, contract, schedule version and threshold set that produced it. And a test in week five that could tell you this recommendation was wrong.